

Soft X-rays And Hard X-rays

Soft X-rays:-

Soft X-rays (often referred to simply as soft rays in specific contexts) occupy the lower energy end of the X-ray spectrum, bridging the gap between extreme ultraviolet (EUV) light and hard X-rays.

They are generally defined as having photon energies between 0.1 keV and 12 keV (100 eV to 12,000 eV), corresponding to wavelengths of approximately 0.1 to 10 nanometers.

Hard X-rays:-

Hard X-rays represent the high-energy end of the X-ray spectrum, typically defined as having photon energies between 10 keV and 120 keV (wavelengths < 0.12 nm).

While the boundary between soft and hard X-rays is not universally fixed (sometimes placed at 5–10 keV), hard X-rays are distinctively characterized by their high penetrating power.

Difference between soft and hard X-rays:

Energy Range: Soft X-rays have lower photon energies, typically

0.1 keV to 12 keV, whereas hard X-rays possess higher energies, generally >10 keV up to 120 keV (or higher).

Wavelength: Soft X-rays have longer wavelengths (approx. 0.1–10 nm), while hard X-rays have shorter wavelengths (approx. <0.12 nm).

Penetration Power: Soft X-rays have low penetrating ability and are easily absorbed by air and light elements (penetrating only nanometers to micrometers in solids). Hard X-rays have high penetrating power, capable of passing through millimeters to centimeters of dense materials like bone and metal.

Interaction with Matter: Soft X-rays interact primarily via photoelectric absorption with light elements (C, N, O), making them surface-sensitive. Hard X-rays interact via Compton scattering and penetrate bulk matter, making them suitable for imaging internal structures.

Experimental Environment: Observations of soft X-rays typically require a vacuum or helium path due to strong air absorption. Hard X-ray experiments can often be conducted in ambient air.

Solar Flares

In solar physics, the relationship between soft and hard X-ray emissions helps diagnose flare dynamics. Observations from

satellites like Yohkoh and RHESSI show that:

Spatial Distribution: Soft X-rays often originate from extended, hot loop structures, while hard X-rays come from more compact footpoints where accelerated electrons impact the denser chromosphere.

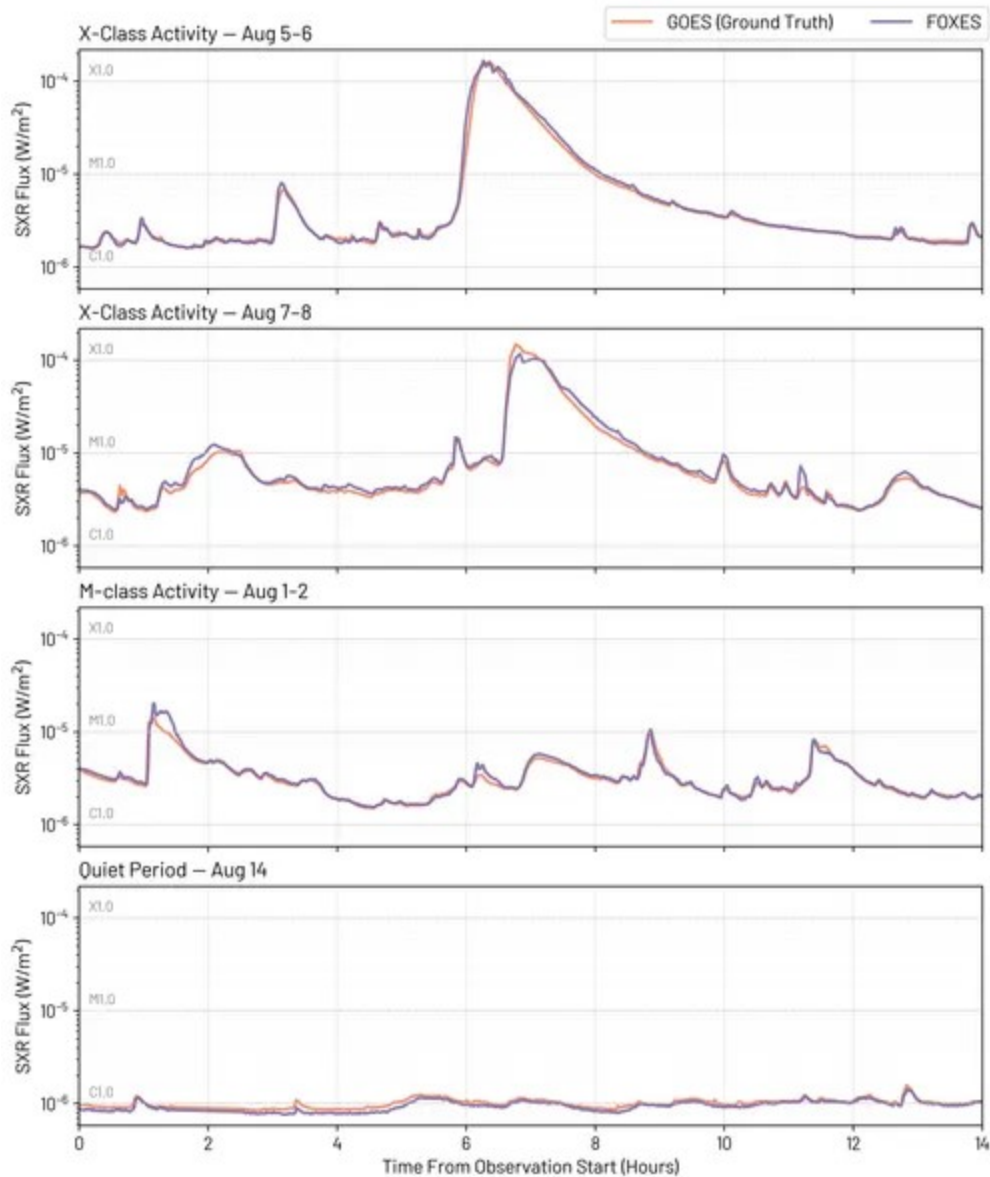
Temporal Correlation: The "Neupert effect" describes a relationship where the time derivative of the soft X-ray light curve is proportional to the hard X-ray time profile. This suggests that the energy deposited by non-thermal electrons (hard X-rays) heats the plasma, which then radiates in soft X-rays.

However, discrepancies exist; some flares show soft X-ray peaks occurring before hard X-ray peaks, indicating complex heating mechanisms.

Graphical Differences in Solar Flares

In solar flare observations, soft X-rays and hard X-rays display distinct graphical signatures in both light curves (intensity vs. time) and spatial maps.

Soft X-rays:



Y-Axis: Soft X-ray flux in Watts per square meter (W/m^2), usually on a logarithmic scale.

X-Axis: Time (UTC).

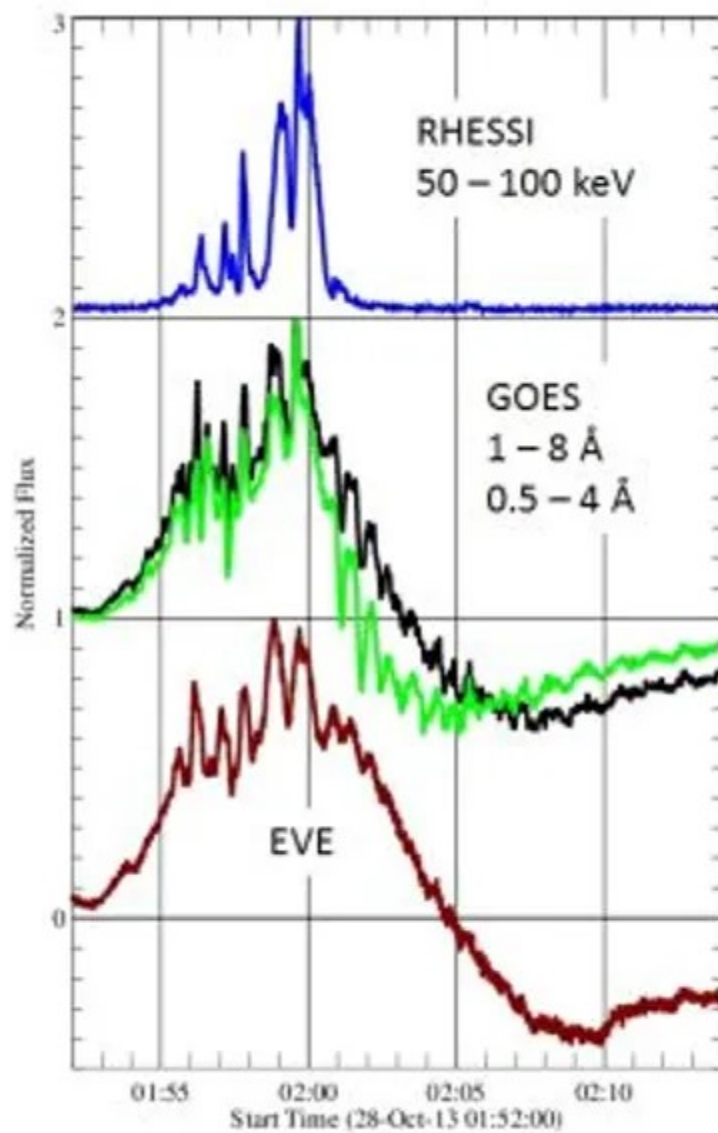
Key Features:

- **GOES Classification:** The vertical scale is divided into classes

A, B, C, M, and X, where each step represents a 10-fold increase in energy (e.g., X1 is 10^{-4} W/m²).

- **Profile:** Solar flares appear as a rapid rise followed by a slower, exponential decay (the "gradual phase").
- **Current Data:** As of June 17, 2026, real-time graphs from GOES-16/18 show flux levels fluctuating typically between A and B class during quiet periods, with sharp spikes during flare events.
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Hard X-rays:



Y-Axis: Count rate (counts/s) or Flux (photons $\text{cm}^{-2} \text{s}^{-1}$), often linear or logarithmic.

X-Axis: Time (UTC).

Key Features:

- **Impulsive Spikes:** Unlike the smooth curves of soft X-rays, hard X-ray graphs (e.g., from RHESSI or STIX) show sharp, narrow spikes lasting seconds. These represent bursts of non-thermal electrons.
- **Energy Bands:** Graphs often overlay multiple energy channels (e.g., 25–50 keV, 50–100 keV, >100 keV). Higher energy bands usually show sharper, shorter-duration spikes.
- **Neupert Effect:** When plotted against soft X-rays, the hard X-ray spikes align with the steepest rise of the soft X-ray curve, not its peak.

Visual graphical differences between soft and hard X-rays:

- **Gradual Smooth Curves:** Unlike the sharp spikes of hard X-rays, soft X-ray light curves (e.g., from GOES) show a smooth, broad profile with a gradual rise and a slower exponential decay. This shape represents the accumulation and cooling of thermal plasma (heat) in the solar corona, rather than instantaneous particle bursts.
- **Temperature Bands:** Graphs often overlay two specific energy channels to derive plasma temperature: 0.5–4 Å (hotter, ~10+ MK) and 1–8 Å (cooler, ~5–10 MK). The ratio between these two bands determines the isothermal temperature of the flare; unlike hard X-rays, higher energy

soft bands do not necessarily show shorter spikes but rather track the heating and cooling phases of the bulk plasma.

- **Neupert Effect Relationship:** When plotted against hard X-rays, the soft X-ray curve acts as the time integral of the hard X-ray bursts. The peak of the soft X-ray curve occurs after the hard X-ray spikes have ceased, corresponding to the moment of maximum accumulated thermal energy, while its steepest rise aligns with the hard X-ray peaks.